

Useful Memory Has a Horizon

Temporal Validity and Structural Limits in Finite-Memory
Distribution Tracking under Drift

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Abstract

Under drift, additional memory reduces finite-sample noise but also increases the age of the evidence used to estimate the present. Finite-memory distribution tracking is therefore governed by a temporal-validity horizon: how long does retained evidence remain statistically useful for representing the current target distribution? The paper studies that question through a general horizon law that separates finite-memory tracking error into a finite-sample term, determined by the chosen geometry and estimator, and a drift-induced staleness term, determined by the regularity of the target path:

$$\mathbb{E} d\left(\widehat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H.$$

Here $a > 0$ is the finite-sample rate exponent and $H \in (0, 1]$ is the temporal roughness exponent. Optimizing this balance yields the temporal-validity horizon $n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$.

A tractable one-dimensional bounded-support fixed-span triangular-array model provides a rigorous root- n finite-sample regime through quantile/Bahadur structure. A Gaussian lower-bound construction shows that the resulting horizon is structural rather than an artifact of a particular estimator. The same object also admits a Gaussian location minimax benchmark on deterministic Hölder paths. For fixed- ε Sinkhorn geometry, the manuscript isolates the closed structural ingredients and states the corresponding horizon inheritance as an explicit conjecture on an embedded fixed-span model.

Memory under drift is governed by temporal validity, not by a universal window rule. The horizon depends jointly on finite-sample geometry and temporal roughness, and retained evidence can become invalid for the present before change becomes statistically detectable.

Keywords: tracking under drift; temporal validity; finite-memory estimation; distribution tracking; Wasserstein distance; Sinkhorn divergence.

MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer represent the present. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The paper studies the temporal-validity horizon of finite memory: how long retained evidence remains useful for estimating the present before drift-induced misalignment overtakes the remaining variance gain.

A general horizon law captures this problem. The starting point is an abstract decomposition

$$\mathbb{E} d\left(\widehat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample rate exponent of the chosen geometry and estimator, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Optimizing this balance yields the temporal-validity horizon

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The horizon depends jointly on finite-sample geometry and on the temporal regularity with which drift accumulates staleness.

The paper develops this object in four steps.

First, it establishes the law in a tractable one-dimensional bounded-support fixed-span triangular-array model. In that regime, the quantile representation of W_2^2 , a triangular-array Bahadur expansion, and a design-effect comparison yield the root- n finite-sample rate

$$\mathbb{E} W_2\left(\widehat{P}_n^{\text{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

This gives a rigorous first proof model for the general horizon law.

Second, it proves a structural lower bound showing that the horizon is imposed by the drift geometry itself rather than by one particular estimator. A refined Gaussian lower-bound construction sharpens the constant-level geometry, and in the Gaussian location model the same object yields a full minimax benchmark on deterministic Hölder paths.

Third, it isolates the closed structural ingredients for fixed- ε Sinkhorn geometry on an embedded fixed-span model and states the corresponding triangular-array horizon inheritance as an explicit conjecture.

Fourth, it formulates the broader family-level inheritance question: which regular distribution families and operational geometries support which finite-sample rates, and therefore which members of the horizon family. Memory under drift is governed by the interaction between finite-sample geometry and temporal roughness, not by a single universal window prescription.

The empirical section mirrors that structure. It makes the finite-memory U-curve visible, shows roughness-dependent horizon scaling, reports evidence relevant to the fixed- ε Sinkhorn conjecture, and exhibits a measurable lag between validity loss and

the first detector alarm. Temporal validity and changepoint evidence are different statistical objects: retained evidence can stop representing the present before change becomes detectable.

This framing is adjacent to several literatures—dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry—but the paper asks a different question. Dynamic-regret analysis asks how well one tracks a moving comparator, and adaptive-window or drift-detection methods ask when the accumulated evidence is sufficient to react. The question here is different: for finite-memory tracking under drift, how long does retained evidence remain temporally valid for estimating the present distribution before age becomes the dominant source of error?

The paper proceeds as follows. Section 2 develops the general horizon law, drift-regularity bounds, a tractable proof model, a structural lower bound, and the benchmark and operational regimes. Section 3 presents the corresponding empirical signatures. Section 4 discusses related work, and Section 5 discusses scope, limitations, and open problems.

2 A Horizon Law for Finite-Memory Tracking under Drift

This section fixes the finite-memory objects and develops the horizon law, structural lower bound, benchmark theorem, and conjectural extensions used in the rest of the paper.

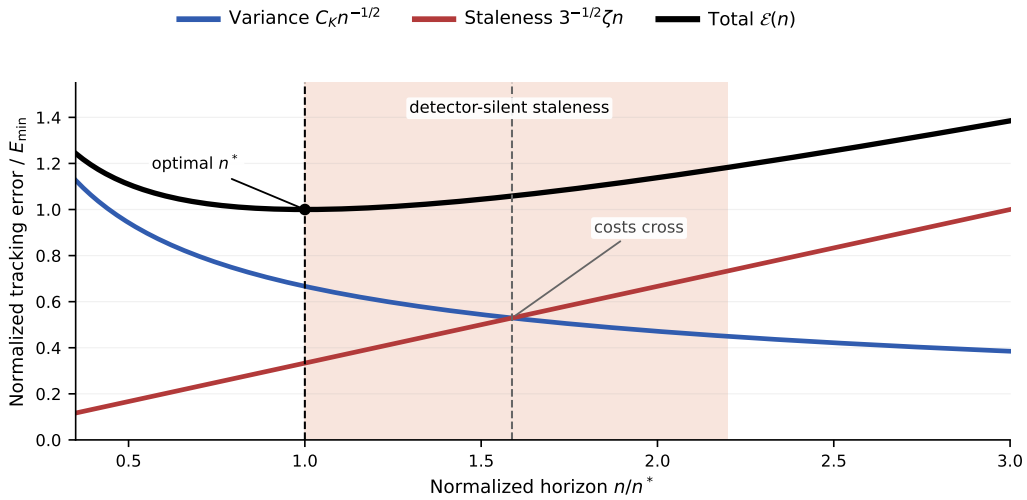


Figure 1: Two clocks of drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^* . The shaded region marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

For a finite-memory estimator with lag weights $w_0, \dots, w_{n-1}$, the natural age

variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j w_j.$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j W_2(P_{t-j}, P_t).$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants.

Throughout the paper, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\hat{P}_t^{(n)}$ the estimator built from the last n observations. The finite-sample term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t .

The theory combines three ingredients: a finite-sample bound relative to $\bar{P}_t^{(n)}$, a path-class control on the staleness between $\bar{P}_t^{(n)}$ and P_t , and, when available, a lower-bound or benchmark theorem showing that the resulting horizon is structural on the geometry under study.

2.1 General Horizon Law

The main theorem separates finite-sample behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law). *Let d be a probability metric satisfying the triangle inequality. At time t , let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E} d\left(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \leq C_K n^{-a}$$

and

$$d\left(\bar{P}_t^{(n)}, P_t\right) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E} d\left(\hat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\left(\hat{P}_t^{(n)}, P_t\right) \leq d\left(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + d\left(\bar{P}_t^{(n)}, P_t\right).$$

Taking expectations and applying the two assumptions yields the claim. $\square$

The proof is immediate, but the theorem isolates the structure of the problem: the path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Corollary 2.2 (Optimized temporal-validity horizon). *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a , H , and C_S , at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding optimized error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimized error scale. $\square$

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.2 Drift Regularity and Staleness Accumulation

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \{(P_s)_{s \in \mathbb{Z}} : W_2(P_{t-j}, P_t) \leq \zeta j^H \text{ for all } j \geq 0\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.3 (Hölder staleness bound via averaged couplings). *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore

$$W_2(\bar{P}_t^{(n)}, P_t) \leq \zeta \left(\frac{1}{n} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2} = C_{H,n} \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2}.$$

In particular, $C_{H,n} \rightarrow (2H+1)^{-1/2}$ as $n \rightarrow \infty$.

Proof. For each lag j , let π_j be an optimal coupling between P_{t-j} and P_t . Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t , so

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \int \|x - y\|^2 d\bar{\pi}(x, y) = \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t).$$

Applying the Hölder envelope gives the displayed finite- n constant, and taking square roots completes the proof. $\square$

Combining lemma 2.3 with corollary 2.2 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root- n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite- n constant in lemma 2.3 is the constant used throughout the paper for uniform windows; the simpler $(2H + 1)^{-1/2}$ expression is its large-window limit.

2.3 A Tractable One-Dimensional Proof Model

The abstract law needs at least one rigorous finite-sample instantiation. The bounded-support fixed-span one-dimensional triangular array provides a tractable proof model for that role.

Proposition 2.4 (Root- n finite-sample bound in the 1D proof model). *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^n P_{n,j}.$$

Assume:

- (i) *all $P_{n,j}$ are supported on a common compact interval $[L, U]$;*
- (ii) *the within-window drift span is uniformly bounded in n ;*
- (iii) *each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;*
- (iv) *writing q_n for the quantile function of $\bar{P}_n$ and $\hat{q}_n$ for the empirical quantile function of $\hat{P}_n^{\text{tri}}$, one has the Bahadur representation*

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \quad u \in (0, 1),$$

with

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

Then

$$\mathbb{E} W_2^2(\widehat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}),$$

and therefore

$$\mathbb{E} W_2(\widehat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2(\widehat{P}_n^{\text{tri}}, \bar{P}_n) = \int_0^1 (\widehat{q}_n(u) - q_n(u))^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\widehat{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \widehat{F}_n(q_n(u))}{\widehat{f}_n(q_n(u))}.$$

Then

$$\mathbb{E} \int_0^1 (\widehat{q}_n(u) - q_n(u))^2 du = \mathbb{E} \int_0^1 (Z_n(u) + R_n(u))^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq x\},$$

independence across the row gives

$$\text{Var}(\widehat{F}_n(q_n(u))) = \frac{1}{n^2} \sum_{j=1}^n p_{n,j}(u)(1 - p_{n,j}(u)) \leq \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E} \int_0^1 Z_n(u)^2 du \leq m^{-2} \int_0^1 \text{Var}(\widehat{F}_n(q_n(u))) du \leq \frac{1}{4m^2n}.$$

By item (iv),

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is controlled by Cauchy–Schwarz:

$$\mathbb{E} \left| 2 \int_0^1 Z_n(u) R_n(u) du \right| \leq 2 \left(\mathbb{E} \int_0^1 Z_n(u)^2 du \right)^{1/2} \left(\mathbb{E} \int_0^1 R_n(u)^2 du \right)^{1/2} = o(n^{-1}).$$

Hence

$$\mathbb{E} W_2^2(\widehat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}).$$

Jensen's inequality then gives

$$\mathbb{E} W_2(\widehat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root- n . $\square$

Remark 2.5 (Sufficient conditions for the proof model). The only nontrivial assumption in proposition 2.4 is item (iv). For i.i.d. quantile processes, Bahadur representations of this type are classical; see Bahadur [1966] and van der Vaart [1998, Chapter 21]. The present paper uses the triangular-array version as an explicit hypothesis for the fixed-span proof model rather than claiming a general theorem that verifies it. This covers, for example, bounded-support smooth location perturbations of a common base density and bounded smooth monotone transforms of such a base law, provided the fixed-span condition is maintained and the stated integrated remainder bound holds.

2.4 Structural Lower Bound

The upper law would be substantially weaker if it were only the optimum of one estimator. The lower bound shows that, in the root- n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

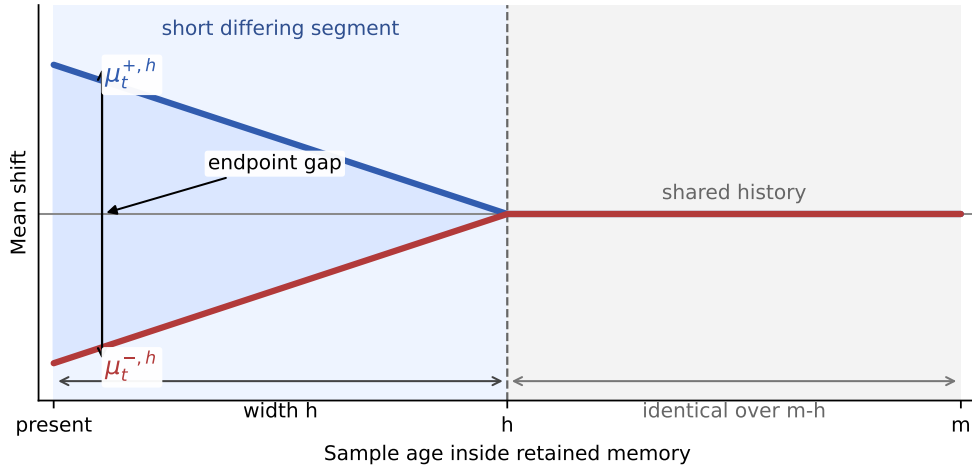


Figure 2: Gaussian lower-bound construction. The two paths share most of the retained history and differ only on a recent segment of width h , yet remain separated at the present. Optimizing that recent segment already produces the same horizon exponent as the upper law in the root- n regime.

Proposition 2.6 (Structural lower bound in the root- n regime). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \quad j = 0, 1, \dots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\text{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\hat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}^{\text{Gauss}}} \mathbb{E} W_2(\hat{P}_t, P_t) \geq c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root- n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof outline. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H . On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^h r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min \left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^h r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H . Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. The exact Gaussian two-point calculation used here is recorded in appendix A.

The lower bound is subclass based rather than class-tight, but that is enough for the central claim: the horizon is a structural feature of the problem rather than an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes a sharper constant-level geometry. The next proposition records the best constant-level refinement presently available within that proof scheme.

2.5 Refined Gaussian Lower-Bound Geometry

Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.7 (Refined Gaussian lower-bound constants). *Fix $H \in (0, 1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H} \Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp profile $\mu_{t-j}^{\pm,h} = \pm \zeta(h-j)_+^H$, let

$$L_h^{\text{ramp}}(\sigma, \zeta) := \zeta h^H \Phi \left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h r^{2H}} \right).$$

Then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\text{ramp}} = (2H + 1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\min} = h^H - (h - r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 (1 - (1 - u)^H)^2 du = \frac{2H^2}{(H + 1)(2H + 1)}.$$

If

$$L_h^{\min}(\sigma, \zeta) := \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h (g_r^{\min})^2}\right),$$

then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For $H < 1$, this strictly improves the ramp constant by the factor

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{H + 1}{2H^2}\right)^{H/(2H+1)} > 1,$$

while at $H = 1$ the two profiles coincide.

Proof. The exact Gaussian testing problem reduces the two-point lower bound to the displayed quantities L_h^{ramp} and $L_h^{\min}$. After rescaling by $h = A(\sigma/\zeta)^{2/(2H+1)}$, the ramp energy and endpoint-minimal energy have limits $(2H + 1)^{-1}$ and I_H , respectively. The corresponding normalized payoff becomes a scalar multiple of $x^{p_H} \Phi(-x)$, whose unique maximizer is x_H . The endpoint-minimal profile minimizes testing energy among all endpoint-saturating Hölder profiles, so it yields the sharper constant. Full details are given in appendix A. $\square$

These refinements sharpen the constant-level lower-bound picture, but the main theoretical role of this section is narrower: they strengthen the Gaussian construction without changing the horizon exponents or the status of the main results.

The same refined lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.8 (Gaussian location minimax benchmark on the deterministic Hölder class). *Fix $H \in (0, 1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s \in \mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \quad \text{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \quad j = 0, 1, \dots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimized finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n \geq 1} \inf_{\hat{P}_t^{(n)}} \sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H , such that

$$c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} \leq \mathcal{R}_H^*(\sigma, \zeta) \leq C_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H(2\Phi(y_H) - 1) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\text{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\text{mean}} = ((H+1)y_H)^{-1/(2H+1)} \left(2\varphi(y_H) + y_H(2\Phi(y_H) - 1) \right).$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\hat{\mu}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} Y_{t-j}$$

and set $\hat{P}_t^{(n)} = N(\hat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2(N(m_1, \sigma^2), N(m_2, \sigma^2)) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n} \sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t) \leq \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n} \sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for

y_H , and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta(h^H - j^H)_+, \quad j \geq 0,$$

with $\beta \leq \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta \min(h^H, j^H) \leq \zeta j^H.$$

Moreover, for fixed endpoint value βh^H , it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the refined lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Optimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem. $\square$

2.6 Benchmark and Operational Regimes

The one-dimensional proof model is the first rigorous finite-sample instantiation. Two broader settings matter for scope: a Gaussian benchmark in which the law is closed on a full path class, and an operational Sinkhorn regime in which some structural ingredients are closed but the full inheritance theorem remains conjectural.

Operational regime. For fixed- ε debiased Sinkhorn, the question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The intended theorem shape is

$$\mathbb{E} S_\varepsilon(\widehat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}) \leq C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed- ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark. The manuscript does not prove that theorem in full generality. What is closed is the structural part of the setup. What remains open is the full triangular-array inheritance statement.

Definition 2.9 (Fixed- ε operational setting). Fix a bounded embedded fixed-span model, an intrinsic complexity index k , a dual Hölder smoothness level α , and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed- ε operational setting if:

- (i) the i.i.d. fixed- ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;
- (ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;

(iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.10 (Structural ingredients for fixed- ε Sinkhorn). *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.9 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Conjecture 2.11 (Fixed- ε Sinkhorn horizon inheritance). *Assume the fixed- ε operational setting from definition 2.9. Then, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the triangular-array finite-sample term should satisfy*

$$\mathbb{E} S_\varepsilon \left(\widehat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)} \right) \leq C_\varepsilon n^{-1/2}, \quad \varepsilon \in \mathcal{B},$$

and the corresponding horizon scale should satisfy

$$n_\varepsilon^*(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε , then the operational horizon law should be

$$n_\varepsilon^*(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Evidence. The i.i.d. benchmark comes from existing fixed- ε entropic transport results [Stromme, 2023, Groppe and Hundrieser, 2024]. In the present embedded fixed-span model, support-complexity inheritance is exact and the dual smoothness threshold reduces to $\alpha > k/2$. Calibration supplies the only remaining ingredient: evidence that the triangular and i.i.d. rate estimates stay close across the regularization band. That evidence motivates the conjecture, but it does not close it as a theorem.

Regular-family inheritance. The broader extension question is whether regular dominated families inherit the same horizon logic with exponents determined by local geometry. The relevant setting is a family $\{P_\theta : \theta \in \Theta\}$ with locally quadratic testing geometry and metric comparability

$$d(P_\theta, P_{\theta'}) \asymp \|\theta - \theta'\|^\alpha$$

for some metric exponent $\alpha > 0$.

Conjecture 2.12 (Regular-family horizon inheritance). *Let d be a probability geometry and let $\{P_\theta : \theta \in \Theta\}$ be a locally regular dominated family for which the i.i.d. benchmark satisfies a metric finite-sample law*

$$\mathbb{E} d \left(\widehat{P}_{\text{iid}}^{(n)}, P_\theta \right) \leq C_K n^{-a}$$

uniformly on the parameter region of interest. Suppose the path obeys a deterministic Hölder envelope in parameter space,

$$\|\theta_{t-j} - \theta_t\| \leq \zeta j^H.$$

Then the induced metric staleness should scale as

$$d(P_{\theta_{t-j}}, P_{\theta_t}) \lesssim \zeta^\alpha j^{\alpha H},$$

so the corresponding tracking problem should satisfy the horizon-family member

$$n^*(a, \alpha H) \asymp (C_K / \zeta^\alpha)^{1/(a+\alpha H)},$$

with optimized risk of order

$$C_K^{\alpha H/(a+\alpha H)} \zeta^{\alpha a/(a+\alpha H)}.$$

In the locally regular parametric subcase, the natural metric finite-sample exponent is $a = \alpha/2$, giving the horizon scale

$$n^*(\alpha/2, \alpha H) \asymp (C_K / \zeta^\alpha)^{1/(\alpha(1/2+H))}.$$

Regular dominated families are the natural theorem targets. Support-changing and singular families remain genuine obstruction classes, so transport-Hölder control alone is too weak to support a single blanket extension theorem.

Together, these settings show how the abstract law extends beyond the one-dimensional proof model while keeping the formal claims within their proper scope.

Once a finite-sample regime and a roughness level are specified, the law yields an explicit memory scale and an optimized error scale. Rougher drift and weaker finite-sample rates both force shorter memory and larger residual error.

3 Empirical Signatures

The empirical section makes the theory visible in finite samples. Its purpose is to exhibit the main observable signatures of the temporal-validity horizon and to report evidence relevant to the fixed- ε Sinkhorn conjecture.

All experiments in this section are synthetic and fully specified by the code in the repository. The Gaussian U-curve uses 20 Monte Carlo seeds; the roughness-family sweep uses 12 seeds across the stated Hölder envelopes; the fixed- ε operational map is calibrated from log-log finite-sample fits over sample sizes (24, 48, 96, 160) on the ambient/intrinsic pairs (8, 1), (8, 2), (12, 1), and (12, 2) using 24 seeds; and the validity-detection experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page-Hinkley detectors. In the operational map, a cell is marked useful when both fitted finite-sample exponents stay above the usefulness threshold and their gap remains below the calibration tolerance. Reported curves are seed-averaged means, while the map summarizes log-log slope fits across the listed sample sizes.

Four signatures matter. First, finite-memory risk should exhibit an interior optimum. Second, that optimum should move with temporal roughness. Third, the fixed- ε Sinkhorn conjecture should receive structured empirical support rather than a single undifferentiated claim. Fourth, temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 A Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance-staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

This is the basic empirical signature of temporal validity under drift. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence.

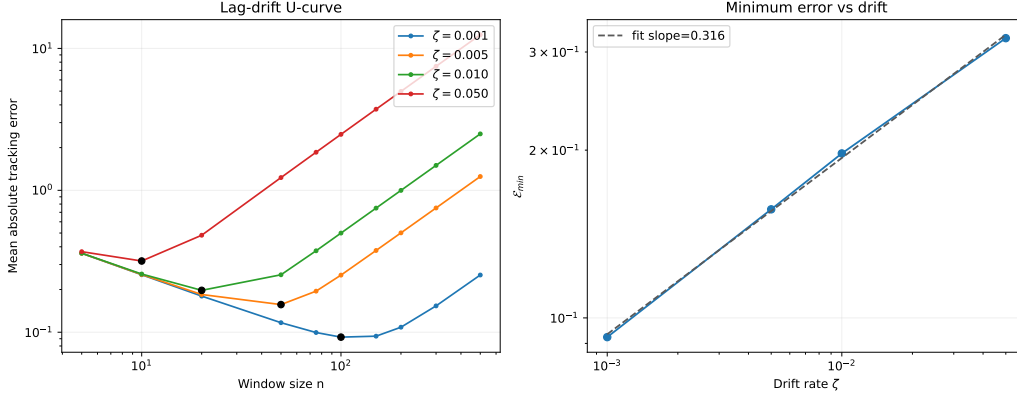


Figure 3: Finite-memory U-curve. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

3.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

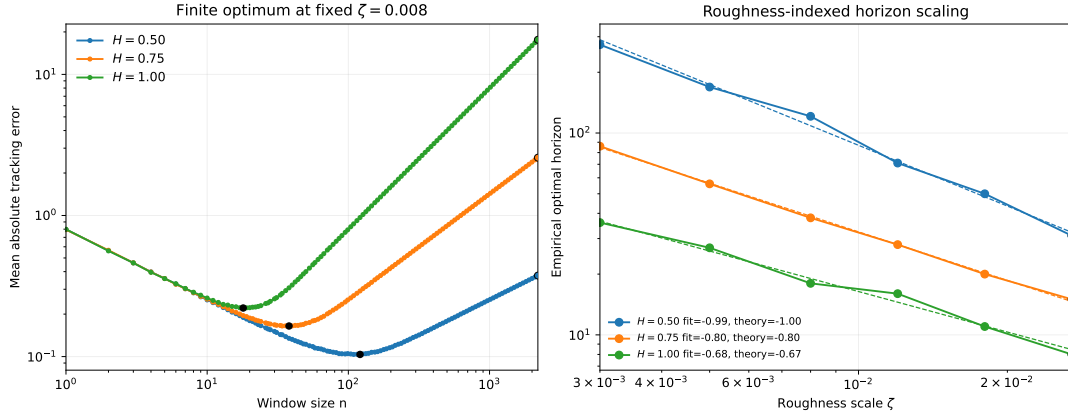


Figure 4: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H . The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

3.3 Evidence for the Sinkhorn Conjecture

The fixed- ε Sinkhorn experiments address a different question: when a practical geometry is used in place of raw W_2 , does the finite-sample term remain stable enough

for the conjectured horizon law to stay informative? The calibrated embedded fixed-span model shows a visible regime map.

The figure below concerns fixed- ε Sinkhorn geometry on embedded low-intrinsic-dimensional support. It shows that the corresponding finite-sample term remains stable across the tested regularization values and that the triangular slope stays close to the i.i.d. benchmark rather than exhibiting a separate exponent. Combined with the structural ingredients isolated in proposition 2.10, this provides empirical support for conjecture 2.11. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the maximal observed bands are $\varepsilon_{\max} = 0.5$ for $(d, k) = (8, 1)$, $(8, 2)$, and $(12, 2)$, and $\varepsilon_{\max} = 0.2$ for $(12, 1)$.

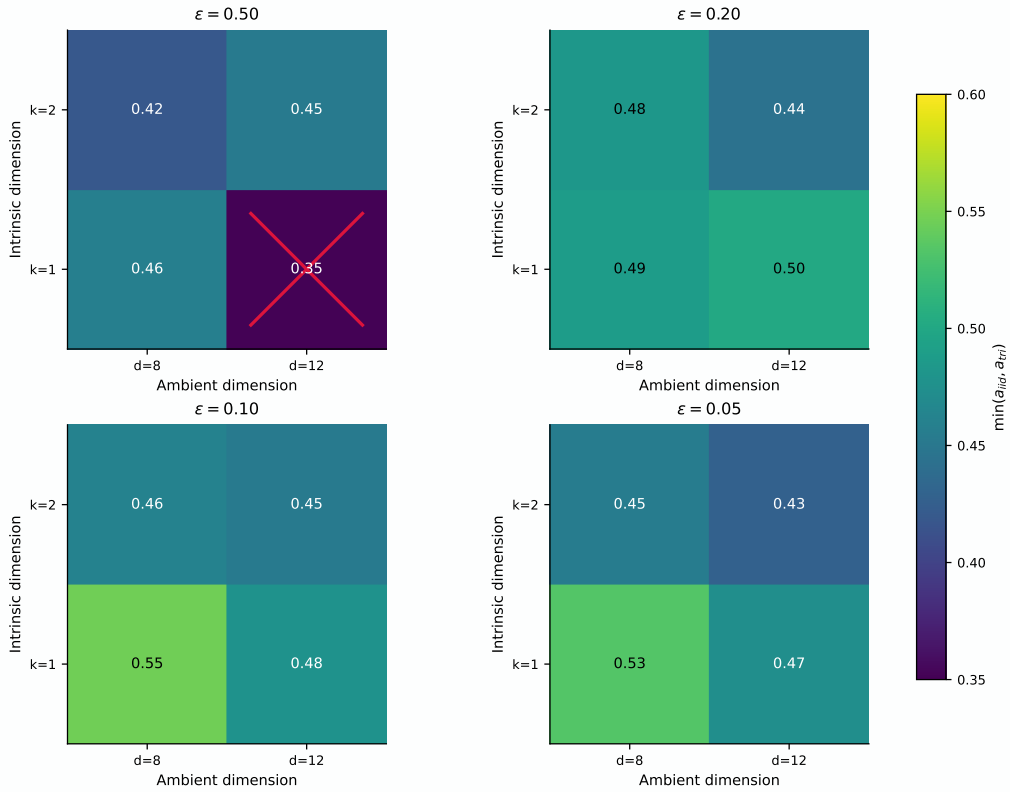


Figure 5: Empirical shape relevant to the fixed- ε Sinkhorn conjecture. Each panel reports the minimum of the i.i.d. and triangular finite-sample exponents over the ambient/intrinsic pairs tested at one regularization level. Crossed cells mark settings that fail the usefulness criterion. The resulting picture has a stable core and a single mixed boundary case on the tested grid.

This map gives the shape of the empirical support behind the conjecture. The observed operational region is neither empty nor uniform across settings; the calibration details and supporting tables are kept in appendices B and C.

3.4 Validity Loss Before Detector Alarm

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over

time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{val-det}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

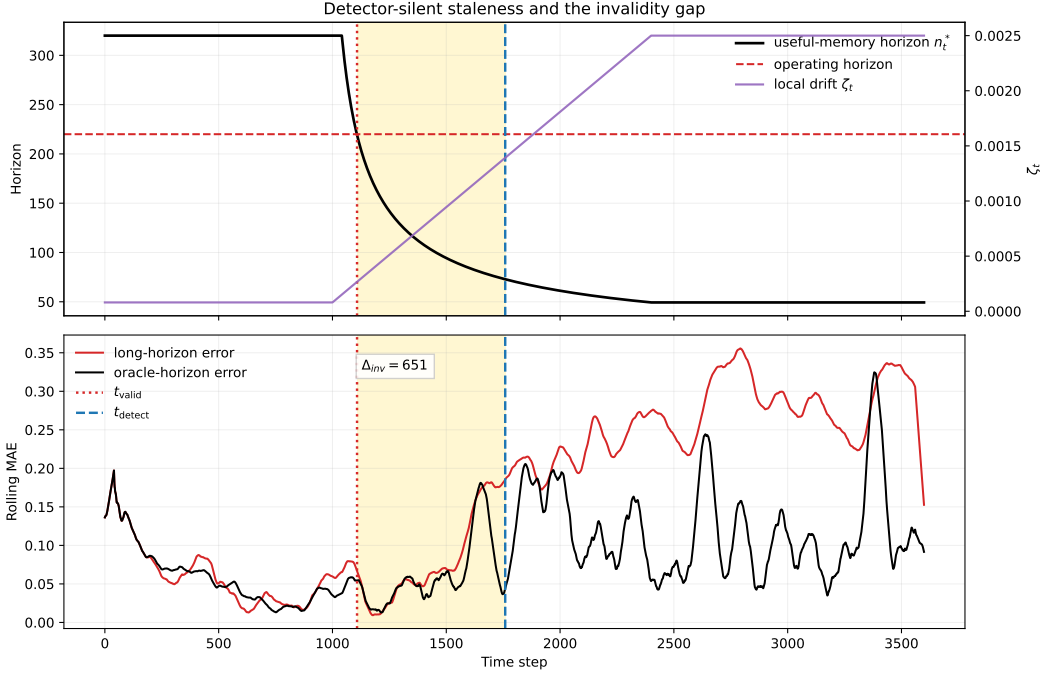


Figure 6: Validity loss before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid} ; the detector responds only later at t_{detect} , leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the same qualitative lag persists under a different detector family as well.

3.5 What the Evidence Shows

Taken together, the experiments support four claims that match the theory. Finite-memory tracking exhibits a structural U-curve rather than an ever-growing memory benefit. The temporal-validity horizon is roughness dependent. The fixed- ϵ Sinkhorn conjecture receives structured empirical support on the tested grid. Temporal validity can fail before changepoint evidence appears, producing a measurable validity-detection lag.

4 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. In that sense, the paper is about the validity horizon of retained evidence rather than the stopping time of a reaction rule. The validity-detection lag makes that distinction visible in finite samples.

Adaptive filtering and state-space tracking. Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The present paper instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

Locally stationary smoothing. Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias–variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Genevay et al., 2019, Mena and Weed, 2019, Stromme, 2023, Groppe and Hundrieser, 2024, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

Entropic and smoothed transport as operational geometries. Regularization changes the statistical geometry as well as the computational one. Entropic and other smoothed transport quantities introduce explicit bias–variance–regularization

trade-offs and can recover more stable statistical behavior than raw high-dimensional Wasserstein distances [Genevay et al., 2019, Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024, Akyildiz et al., 2024]. That perspective is exactly why the paper treats fixed- ε Sinkhorn as an operational geometry rather than as a surrogate for the raw W_2 theorem. Regularization can induce a usable finite-sample regime on which a finite-memory horizon law still makes sense.

Wasserstein nonstationarity and robust optimization. Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in W_1 [Amini and Vinas, 2026]. Those results are adjacent to the finite-sample-regime issue studied here, but they do not by themselves close the present W_2 window-mixture concentration problem. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

Freshness and Age of Information. The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory is governed by a temporal-validity horizon that balances finite-sample accuracy against drift-induced staleness, and temporal validity can fail before change becomes statistically detectable. The resulting viewpoint is complementary to regret and detection analyses: it characterizes when memory itself ceases to be a statistically valid proxy for the present.

5 Limitations and Scope

The paper’s central object is the temporal-validity horizon for finite-memory tracking under drift. The main closed results are the general horizon law, the uniform-window staleness bound, the one-dimensional root- n proof model, the structural Gaussian lower bound at the exponent level, and the Gaussian location minimax benchmark. Other parts of the manuscript are intentionally stated with narrower status.

Finite-sample scope. The theory is modular in the finite-sample exponent a , but the main rigorous instantiation is the root- n regime. The tractable proof model establishes that regime in one dimension under bounded support and fixed span. Outside that setting, the same balancing logic continues to hold, but the finite-sample term must come from a different theorem. In particular, the paper does not claim a general raw- W_2 triangular-array theorem in arbitrary dimension.

Operational scope. For fixed- ε Sinkhorn, the manuscript proves the structural ingredients of the embedded fixed-span model and states the full triangular-array inheritance law as a conjecture. Exact support-complexity inheritance and the dual threshold $\alpha > k/2$ are closed; the remaining step is the full theorem connecting the triangular-array finite-sample term to the i.i.d. benchmark.

Family-level scope. The manuscript does not claim a blanket extension to every deterministic W_2 -Hölder path class. Continuous regular families and operational geometries can support meaningful finite-sample exponents, but support-changing and singular families behave differently. The natural extension target is therefore a theorem for regular dominated families or operational geometries with comparable local testing structure.

Lower-bound scope. The structural lower bound is subclass based. The Gaussian construction is enough to show that, in the root- n regime, the horizon is structural rather than an artifact of a particular estimator. What remains open is class-tightness and constant-sharpness for broader deterministic Hölder path classes and for finite-sample regimes beyond $a = 1/2$. The refined Gaussian geometry sharpens the constants available within that lower-bound program, but it does not close the full distributional lower theory.

Weights and memory shape. The main text emphasizes uniform windows. The paper does not investigate whether broader non-uniform weights can improve constants or alter exponents across richer path classes.

Online roughness estimation. The law is stated in terms of the roughness parameters H and ζ , but the paper does not propose a fully online procedure for estimating them or for quantifying sensitivity to misspecification. Turning the horizon law into a robust adaptive policy remains future work.

Empirical scope. The experiments are synthetic and structurally motivated. Their role is to make the theory visible and to provide evidence for the operational conjecture, not to demonstrate competitive performance on a benchmark suite. Real-data validation and broader comparisons with adaptive horizon-selection rules remain outside the present scope.

Detectability versus validity. Validity loss before detector alarm is an empirical consequence of the theory, not the primary theorem object. The paper shows that temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. The next technical questions are clear. The first is to sharpen the finite-sample constant in the one-dimensional proof model. The second is to understand when practical geometries, such as fixed- ε Sinkhorn, admit full triangular-array inheritance theorems in genuinely mid- or high-dimensional settings. The third is to determine which regular dominated distribution families support which finite-sample exponents and to extend the lower theory accordingly. Beyond that come

class-tight lower bounds, non-uniform weighting, real-data validation, and stronger stochastic-process formulations.

6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes finite memory a temporal-validity object rather than an unlimited statistical resource.

The paper formalizes that idea through the horizon law

$$\mathbb{E} d\left(\widehat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H,$$

whose optimized scale is

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The scientific claim is that finite-memory tracking under drift is governed by this balance between finite-sample accuracy and drift-induced staleness.

A tractable one-dimensional proof model supplies a rigorous root- n finite-sample regime, and a structural Gaussian lower-bound construction shows that the corresponding horizon is not an artifact of one estimator. The same object yields a Gaussian location minimax benchmark on deterministic Hölder paths. For fixed- ε Sinkhorn on an embedded fixed-span model, the paper proves the structural ingredients and states the full inheritance law as a conjecture supported by the reported evidence. These results do not constitute separate stories. They are different manifestations of one temporal-validity problem.

The empirical section makes that object visible through four signatures: a finite-memory U-curve, roughness-dependent horizon scaling, structured evidence for the Sinkhorn conjecture, and a strictly positive lag between validity loss and detector alarm. The resulting message is both theoretical and operational: retained evidence has a path-dependent horizon, and statistical detectability need not coincide with temporal validity.

The main open questions concern sharper finite-sample constants in the proof model, lower theory beyond the root- n regime, broader family-level lower bounds, and theorem-level operational results for the conjectural Sinkhorn extension. Those questions extend the same object. They do not change what the paper is about.

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Proof Details

This appendix records detailed derivations for the main-text results whose role is structural but whose full calculations would interrupt the main line.

A.1 Refined Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^h r^{2H} \quad \text{and} \quad G_h := \sum_{r=1}^h (h^H - (h-r)^H)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\text{ramp}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{S_h}\right)$$

and

$$L_h^{\text{min}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{G_h}\right).$$

Proof of proposition 2.7. Fix $H \in (0, 1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma} \sqrt{S_h} \rightarrow \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\text{ramp}}(\sigma, \zeta)}{\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}} \rightarrow A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0, \infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\text{ramp}} = (2H + 1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \geq g_h - (h - r)^H = h^H - (h - r)^H$$

for every feasible endpoint-saturating profile g . Therefore the profile

$$g_r^{\min} = h^H - (h - r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit

$$h^{-(2H+1)} G_h \rightarrow I_H = \int_0^1 (1 - (1 - u)^H)^2 du = \frac{2H^2}{(H + 1)(2H + 1)}.$$

With the same scaling

$$h = A \left(\frac{\sigma}{\zeta} \right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma} \sqrt{G_h} \rightarrow A^{H+1/2} \sqrt{I_H}.$$

Setting $x = A^{H+1/2} \sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{I_H^{-1}}{2H + 1} \right)^{H/(2H+1)} = \left(\frac{H + 1}{2H^2} \right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$. $\square$

A.2 Two-point lower bound used in proposition 2.6

The structural lower bound in the main text is the exponent-level consequence of the exact Gaussian two-point calculation above. For the ramp pair

$$\mu_{t-j}^{\pm, h} = \pm \zeta (h - j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\text{ramp}}(\sigma, \zeta)$. Optimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H + 1)^{-1}$ for the root- n regime.

B Repository Guide

The repository contains provenance, regeneration scripts, and extended support material that do not belong in the main text. This appendix records where the principal artifacts are traced.

Artifact generation. The code for manuscript figures and tables is in `code/useful_memory_horizon/`. Generated figures, tables, and CSV outputs used by the manuscript are in `artifacts/`. The repository root `README.md` records the regeneration commands.

Empirical provenance. The finite-memory diagnostics and the operational calibration tables are kept out of the main text. The table sources are in `artifacts/tables/gaussian/finite_sample_geometry.tex` and `artifacts/tables/conceptual/carrier_layers.tex`, while the corresponding appendix discussion is in appendix C.

Proof provenance. The theorem statements are in the main manuscript. Detailed derivations that are part of the paper rather than the repository provenance layer are recorded in appendix A.

C Finite-Sample Diagnostics

This appendix collects finite-sample diagnostics that support the theorem line and the conjectural extensions. These checks are not substitutes for theorem statements; they are part of the repository-backed provenance layer summarized in appendix B.

C.1 Finite-Sample Geometry Check

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log-log slope in n . We also probe the non-identically distributed fixed-span window directly in one dimension by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that check the within-window span ζn^H is held fixed, so the experiment targets the one-dimensional proof-model interpretation rather than a large-span regime in which heterogeneity itself can become dominant. Finally, we check the Sinkhorn proxy at a few fixed values of ε .

This table supports the finite-sample interpretation used in the main text. In low dimension, the raw i.i.d. Wasserstein term is compatible with a root- n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in proposition 2.4.

Table 1: Empirical scaling check for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array check compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn proxy is reported at fixed ε . These rows are numerical support for the finite-sample-regime discussion rather than theorem statements.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
W_2 , i.i.d. mixture, $d = 1, H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2 , i.i.d. mixture, $d = 1, H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

C.2 Extended and Operational Regimes

The next table summarizes broader finite-sample evidence beyond the one-dimensional proof model. The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed- ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 2: Summary rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed- ε Sinkhorn across ε on embedded low-intrinsic-dimensional support in ambient dimension eight.

Regime	Setting	Parameter	Rate exponent a
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.4652
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.5722
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.4737
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.5588
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.4792
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, eps=0.50	0.50	0.5599
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, eps=0.50	0.50	0.5706
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, eps=0.20	0.20	0.5466
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, eps=0.20	0.20	0.6150
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, eps=0.10	0.10	0.4646
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, eps=0.10	0.10	0.5060
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, eps=0.05	0.05	0.4339
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, eps=0.05	0.05	0.5124

These rows provide calibration for the theorem statements in the main text. The extended-regime rows support the claim that low intrinsic dimension preserves a rate near the root- n regime without a dramatic triangular-versus-benchmark gap. The

operational-regime rows supply empirical evidence relevant to the fixed- ε operational setting in definition 2.9: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they support the conjectural inheritance statement discussed in conjecture 2.11. What remains open is a full proof of triangular-array inheritance that removes this empirical calibration step.

References

- O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Míguez. Gaussian entropic optimal transport: Schrödinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.
- Arash A. Amini and Luciano Vinas. Wasserstein concentration of empirical measures for dependent data via the method of moments. *CoRR*, abs/2601.07228, 2026. doi: 10.48550/arXiv.2601.07228.
- R. R. Bahadur. A note on quantiles in large samples. *The Annals of Mathematical Statistics*, 37(3):577–580, 1966.
- Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.
- Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007. doi: 10.1137/1.9781611972771.42.
- Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL <https://proceedings.mlr.press/v89/cheung19b.html>.
- Rainer Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997. doi: 10.1214/aos/1034276620.
- Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3–4):707–738, 2015. doi: 10.1007/s00440-014-0583-7.
- João Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Advances in Artificial Intelligence — SBIA 2004*, volume 3171 of *Lecture Notes in Computer Science*, pages 286–295. Springer, 2004. doi: 10.1007/978-3-540-28645-5_29.
- João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):44:1–44:37, 2014. doi: 10.1145/2523813.
- Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.

- Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.
- Michel Groppe and Shayan Hundrieser. Lower complexity adaptation for empirical entropic optimal transport. *CoRR*, abs/2306.13580, 2024. doi: 10.48550/arXiv.2306.13580.
- Fabian Hinder, Valerie Vaquet, and Barbara Hammer. The window dilemma: Why concept drift detection is ill-posed, 2023. Preprint.
- Andrew H. Jazwinski. *Stochastic Processes and Filtering Theory*, volume 64 of *Mathematics in Science and Engineering*. Academic Press, New York, 1970.
- Jiashuo Jiang, Xiaocheng Li, and Jiawei Zhang. Online stochastic optimization with wasserstein based non-stationarity, 2020. URL <https://arxiv.org/abs/2012.06961>.
- Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960. doi: 10.1115/1.3662552.
- Sanjit Kaul, Roy Yates, and Marco Gruteser. Real-time status: How often should one update? In *Proceedings of IEEE INFOCOM*, pages 2731–2735, 2012. doi: 10.1109/INFCOM.2012.6195689.
- Dominic S. T. Keehan, Edward J. Anderson, and Wolfram Wiesemann. Don’t look back in anger: Wasserstein distributionally robust optimization with nonstationary data. *CoRR*, abs/2510.18566, 2025. doi: 10.48550/arXiv.2510.18566.
- Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL <https://proceedings.mlr.press/v202/mazzetto23a.html>.
- Gonzalo Mena and Jonathan Weed. Statistical bounds for entropic optimal transport: Sample complexity and the central limit theorem. *Advances in Neural Information Processing Systems*, 32, 2019.
- Jonathan Niles-Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A): 2620–2648, 2019. doi: 10.3150/18-BEJ1077.
- Austin J. Stromme. Minimum intrinsic dimension scaling for entropic optimal transport. *CoRR*, abs/2306.03398, 2023. doi: 10.48550/arXiv.2306.03398.
- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998.
- Zhiyong Wang, Jize Xie, Yi Chen, John C. S. Lui, and Dongruo Zhou. Variance-dependent regret bounds for non-stationary linear bandits, 2024. URL <https://arxiv.org/abs/2403.10732>.

Roy D. Yates, Yin Sun, D. Richard Brown, Sanjit K. Kaul, Eytan Modiano, and Sennur Ulukus. Age of information: An introduction and survey. *IEEE Journal on Selected Areas in Communications*, 39(5):1183–1210, 2021. doi: 10.1109/JSAC.2021.3064980.

Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.